

If you are ever under a ping flood attack, use a tool like tcpdump to determine where the packets are coming from (or appear to be coming from), then contact your provider with this information. Ping floods can most easily be stopped at the router level or by using a firewall.

- Ping o' Death - The Ping o' Death attack sends ICMP ECHO REQUEST packets that are too large to fit in the kernel data structures intended to store them. Because sending a single, large (65,510 bytes) "ping" packet to many systems will cause them to hang or even crash, this problem was quickly dubbed the "Ping o' Death." This one has long been fixed, and is no longer anything to worry about.
- Teardrop / New Tear - One of the most recent exploits involves a bug present in the IP fragmentation code on Linux and Windows platforms. It is fixed in kernel version 2.0.33, and does not require selecting any kernel compile-time options to utilize the fix. Linux is apparently not vulnerable to the "newtear" exploit.

You can find code for most exploits, and a more in-depth description of how they work, at <http://www.rootshell.com> using their search engine.

VPNs - Virtual Private Networks

VPN's are a way to establish a "virtual" network on top of some already-existing network. This virtual network often is encrypted and passes traffic only to and from some known entities that have joined the network. VPNs are often used to connect someone working at home over the public Internet to an internal company network.

If you are running a Linux masquerading firewall and need to pass MS PPTP (Microsoft's VPN point-to-point product) packets, there is a Linux kernel patch out to do just that. See: ip-masq-vpn.